Tennis and Biomechanics

Author

Cyril is a tennis coach and physical conditioning expert with over 25 years' experience in the tennis coaching and sport sciences field. He is a qualified "Tennis Professor Level 2", the highest tennis coaching qualification in France and has a Master's degree and PhD in Sport Sciences. Cyril´s professional experiences include: private tennis coaching, through which he has trained many high performance national-ITF junior tour standard players in which several were part of the French Elite National Program, Head Tennis Coach responsibilities at various tennis clubs and training centers in France, Coach Education instruction and teaching. In addition, Cyril´s background also includes presenting at various national and international conferences including the 2015 ITF Worldwide Coaches Conference by BNP Paribas. He has also contributed publications in internationally peer reviewed scientific journals, the ITF Coaching and Sport Science Review and for the French Tennis Federation.

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This article series is aimed to provide coaches and players with better understanding of technical performance factors involved in tennis shots and on-court movement.

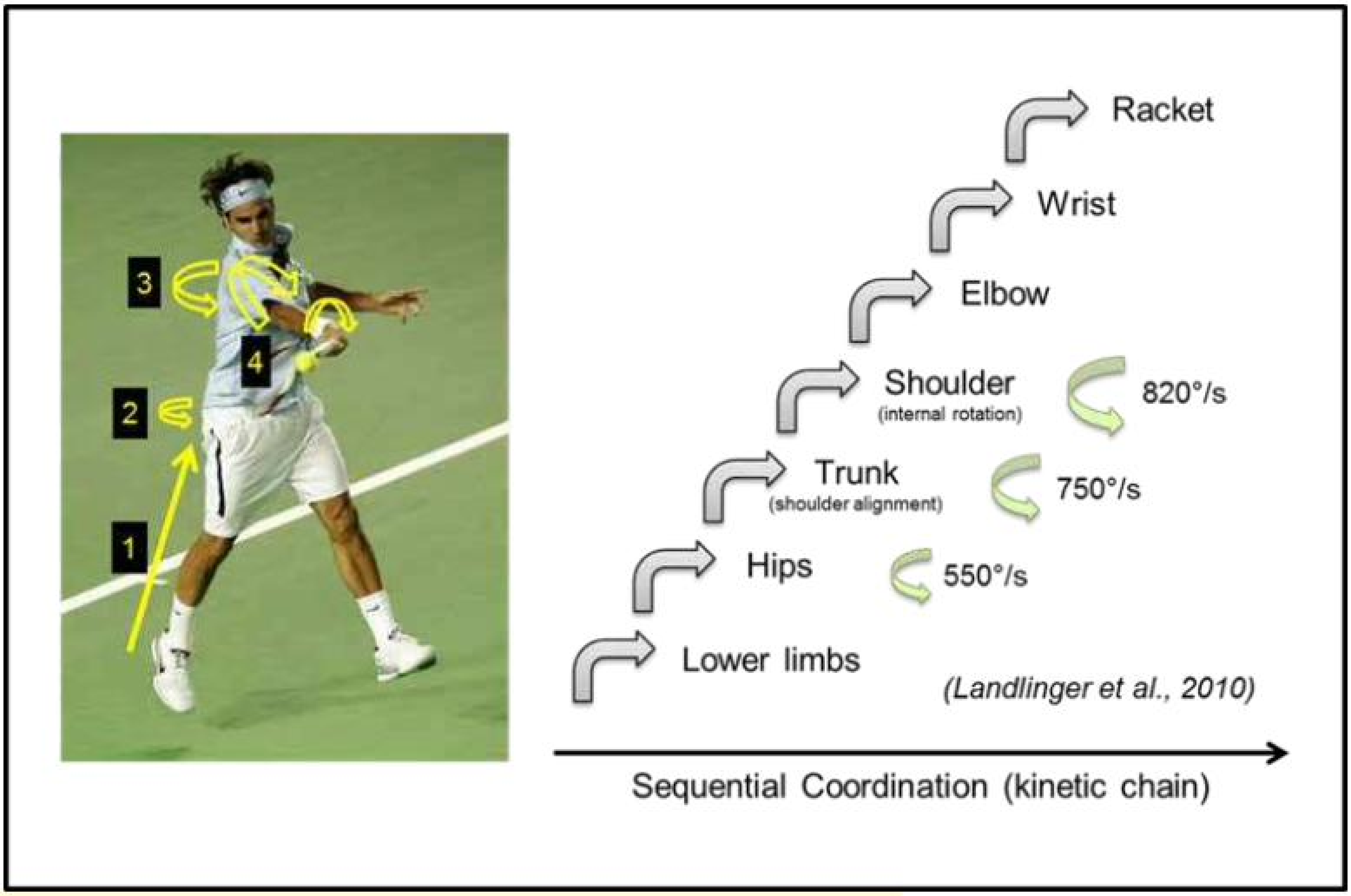
The technical analysis in tennis has long and almost exclusively been a description of gestural form and rhythm from the High level model used as unique reference for the coaches. But this qualitative insight only shows the action's consequences and not the factors generating it (the causes). Then, the technical analysis has become more functional thanks to scientific contributions and especially biomechanics. Through a better understanding of the underlying mechanisms, it is now possible to provide coaches with pertinent visual clues matching the "technical fundamentals" that are absolutely essential for an effective technique performance. The effectiveness is defined, in tennis, as the ability to perform an action with a reduced energetic cost or minimal mechanic stress, and that characterises an "optimum efficiency". This notion of effectiveness is important since it helps to prevent injuries related to the high repetition of the same actions during training sessions and matches.

What is Tennis biomechanics?

It refers to the study of forces and motions and is essentially the science of movement technique involved in tennis. The proper understanding of biomechanics relating to tennis skill has the greatest implications on both performance and injury prevention.

The movement can be analyzed through **quantitative** kinematic, kinetic and electromyographic analyses performed in research settings. Kinematics can be defined as the description of motion without consideration of its causes, including its linear and angular displacements, velocities and accelerations. Kinetics is the study of forces causing the movement and the resultant energy. Electromyography (EMG) is the study of electrical signals related to the muscular contractions.

Scientific studies have highlighted the **"kinetic chain principle"**. The "kinetic chain" term refers to the conceptual framework used



to understand the mechanisms by which athletes perform the necessary complex tasks to function in sport. It relates to the biomechanics through which the necessary forces are generated and that help to regulate and modify the loads imposed on the joints, especially the high loads at the shoulder joint. In a normally functioning kinetic chain, the legs and trunk segments are the engine that develops force while also forming the proximal base for the distal mobility. In simple terms, the energy or force generated by a link (or a body part) can be transferred to the next link in the chain. Moreover, to be effective, this kinetic chain must satisfy the speed summation principle in which a segment initiates its motion when the adjacent proximal segment reaches its maximum angular speed.

If we take the forehand shot as an example, the racket velocity at impact is the product of a summation of forces from the lower limbs up through the trunk and the dominant arm .

The movement can also be analyzed through qualitative analyses of human movement by using six principles of biomechanics for tennis remembered easily by way of the acronym “BIOMEC”:

Balance: ability to maintain equilibrium either dynamically or statically

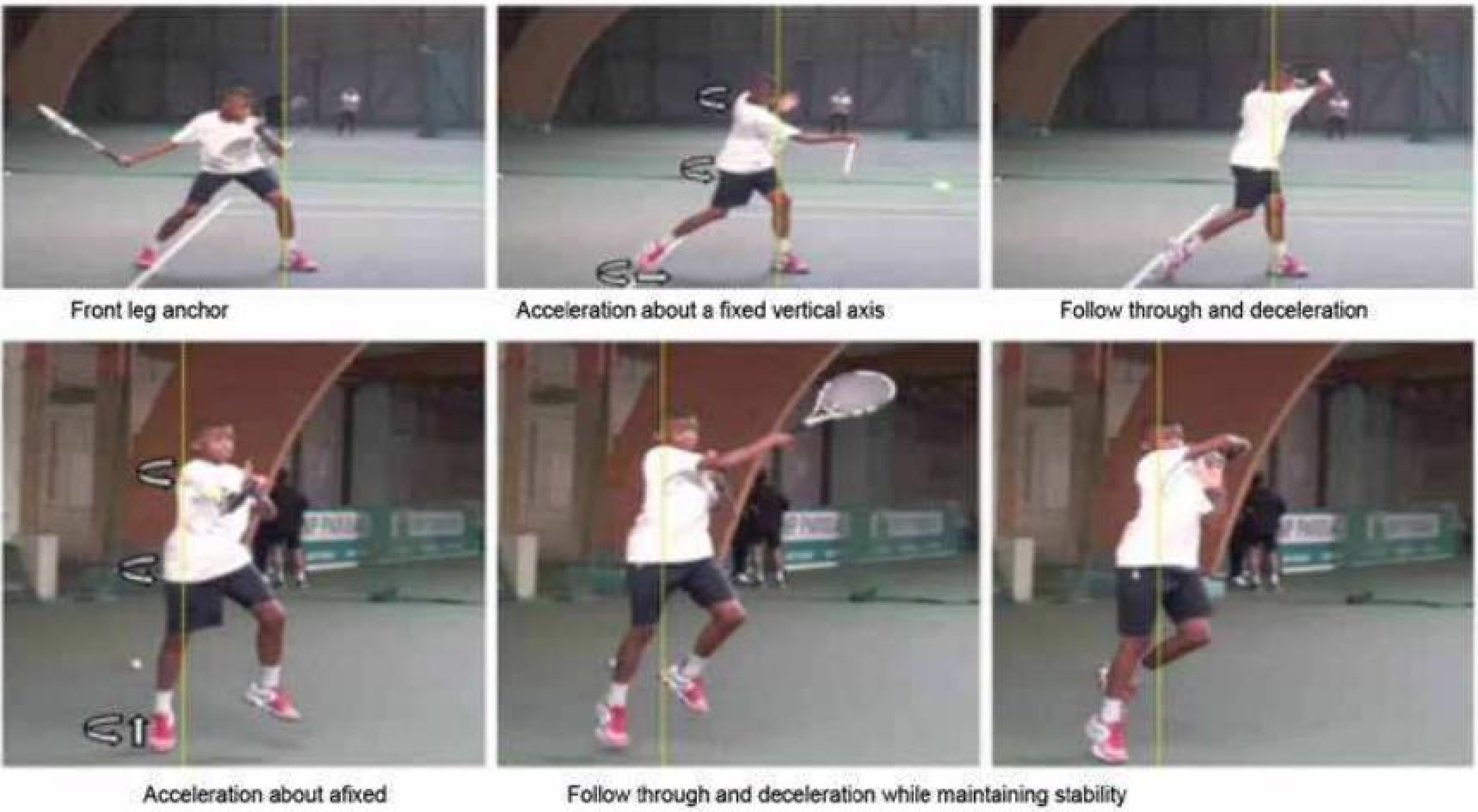
propulsion from the back leg

Anchor and propulsion from the outside leg

Inertia: resistance of a body to move or to stop moving
Opposite force: for every action, there is an equal and opposite reaction
Momentum: amount of mass of the body related to its speed (mass x velocity)
Elastic energy: energy stored in the muscles as a result of stretching the muscle
Coordination: kinetic chain principle
It is the most practical approach for coaches and each of these principles will be addressed in future articles.

Conclusion :

It's important to understand that technical analysis has to be functional and not only descriptive. It allows coaches to work on the causes that produce the optimal conditions of technical performance. The figure below is an example of functional description of forehand shots performed with different stances.
The next article will address the biomechanics principle of opposite force and how it's used by players to move in the tennis court and hit shots more efficiently.



Front leg anchor | Acceleration about a fixed vertical axis | Follow through and deceleration

Acceleration about afixed | Follow through and deceleration while maintaining stability